Normal Form for High-Dimensional h-Cobordisms

1 Goal of the lecture

Let $(W; M_0, M_1)$ be a compact smooth cobordism of dimension n. Thus

$$\partial W = M_0 \sqcup M_1.$$

We say that W is an h-cobordism if both inclusions

$$M_0 \hookrightarrow W, \qquad M_1 \hookrightarrow W$$

are homotopy equivalences. In this lecture we explain why, in dimension at least six, an h-cobordism can be put into a very special handle-theoretic form.

Theorem 1.1 (Normal form lemma)**.** *Let $(W; M_0, M_1)$ be a compact smooth h-cobordism of dimension $n \geq 6$. Let q be an integer satisfying*

$$2 \leq q \leq n - 3.$$

Then there is a handlebody decomposition of W relative to M_0 with only handles of index q and $q+1$. Equivalently, there is a diffeomorphism relative to M_0

$$W \cong M_0 \times [0,1] + \sum_{i=1}^{p_q} h_i^q + \sum_{j=1}^{p_{q+1}} h_j^{q+1},$$

where h_i^r denotes a handle of index r.

This result is often called the normal form lemma. It is the geometric step that makes the later algebra possible. Once the handle decomposition has only two adjacent dimensions, the relative cellular chain complex of $(\widetilde{W}, \widetilde{M}_0)$ has only one possibly non-zero differential. That differential is represented by a square matrix over the group ring $\mathbb{Z}[\pi_1 M_0]$. This matrix is where Whitehead torsion enters the s-cobordism theorem.

For the purposes of this lecture, the proof should be understood through the following slogan:

a low-index handle can be removed at the cost of creating a higher-index handle.

Repeating this operation pushes handles upward until none remain below the chosen index. Then we reverse the cobordism and apply the same idea to the dual decomposition, which pushes the remaining high-index handles downward.

2 Handlebody decompositions relative to one boundary component

A handle of index r in an n-dimensional cobordism is

$$h^r = D^r \times D^{n-r}.$$

It is attached to a boundary component along

$$S^{r-1} \times D^{n-r} \subset \partial(D^r \times D^{n-r}).$$

The disk

$$D^r \times \{0\}$$

is called the core of the handle, and the disk

$$\{0\} \times D^{n-r}$$

is called the cocore. The boundary of the cocore,

$$\{0\} \times S^{n-r-1},$$

is called the belt sphere or transverse sphere of the handle.

A handlebody decomposition of W relative to M_0 is a presentation

$$W \cong M_0 \times [0,1] + \sum_{i=1}^{p_0} h_i^0 + \sum_{i=1}^{p_1} h_i^1 + \cdots + \sum_{i=1}^{p_n} h_i^n$$

relative to $M_0 = M_0 \times \{0\}$. We write W_r for the partial cobordism after all handles of index at most r have been attached:

$$W_r := M_0 \times [0,1] + \sum_{i=1}^{p_0} h_i^0 + \cdots + \sum_{i=1}^{p_r} h_i^r.$$

The boundary component that is changing as we attach handles will be denoted by $\partial_1 W_r$. More precisely, $\partial_1 W_r$ is the part of ∂W_r not equal to the old boundary $M_0 \times \{0\}$.

The geometric meaning of the normal form lemma is that a complicated handlebody decomposition can be simplified until only the middle part remains. However, handles cannot usually be deleted one by one. They are removed by cancelling pairs of consecutive indices.

3 The cancellation lemma

The basic cancellation criterion is the following.

Lemma 3.1 (Handle cancellation)**.** *Let V be an n-dimensional compact cobordism. Suppose that a q-handle h^q has been attached to V, and then a $(q+1)$-handle h^{q+1} is attached. If the attaching sphere $S^q \times \{0\}$ of h^{q+1} meets the transverse sphere of h^q transversely in exactly one point, then the two handles cancel. In other words,*

$$V + h^q + h^{q+1} \cong V$$

relative to the incoming boundary.

The condition "one transverse intersection point" is the smooth analogue of an elementary collapse in CW theory. The q-handle and the $(q+1)$-handle are attached in the simplest possible non-trivial way, and together they contribute nothing to the diffeomorphism type of the cobordism.

In a handlebody decomposition with many q-handles, the attaching sphere of a new $(q+1)$-handle must meet the transverse sphere of the particular q-handle we want to cancel exactly once and must be disjoint from the transverse spheres of all the other q-handles. Much of the proof is devoted to arranging exactly this situation.

Remark 3.2. *The cancellation lemma is a local statement, but arranging its hypothesis is global. The attaching sphere may wind around the cobordism, and its algebraic intersections with belt spheres are measured in the group ring $\mathbb{Z}[\pi_1 W]$. This is why a purely visual proof is not enough.*

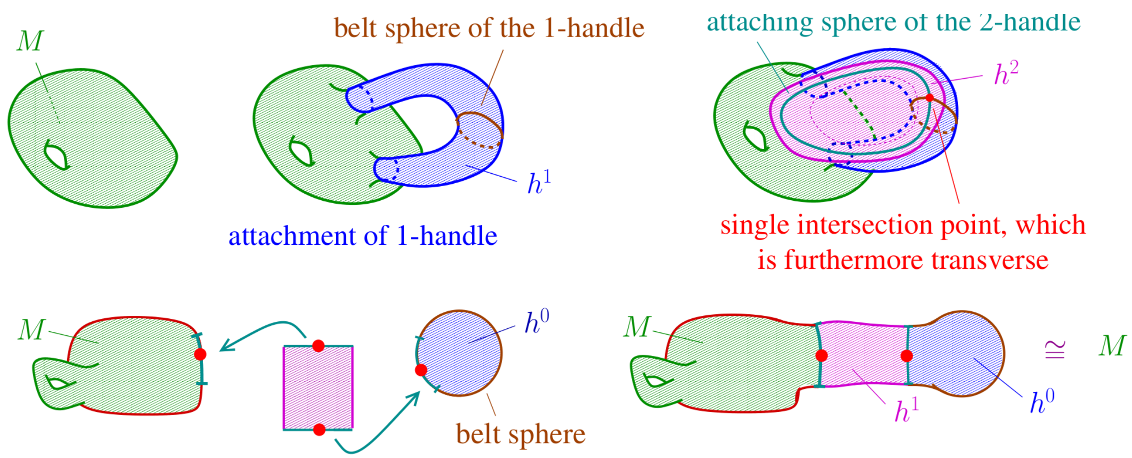

Figure 1: Two basic cancellation models from Friedl's notes. The first picture shows a 1–2 cancellation in dimension three: the attaching sphere of the 2-handle meets the belt sphere of the 1-handle in one transverse point. The second picture shows the 0–1 cancellation model.

Example 3.3 (Handle cancellation examples from the reference)**.** *Friedl's Chapter 256 gives several useful models to keep in mind.*

(1) ***A** 1–2 **cancellation in dimension three.*** *Start with a 3-manifold M, attach a 1-handle, and then attach a 2-handle whose attaching circle meets the belt sphere of the 1-handle transversely in exactly one point. The cancellation theorem gives*

$$M \cup h^1 \cup h^2 \cong M.$$

This is the most concrete picture of the criterion: the attaching circle of the 2-handle crosses the belt sphere of the 1-handle once.

(2) ***The** 0–1 **case.*** *Attaching a 0-handle gives $M \cup h^0 = M \sqcup B^n$. If a subsequent 1-handle has one attaching foot on ∂M and the other on $\partial B^n = S^{n-1}$, then its attaching 0-sphere meets the belt sphere of the 0-handle once. Hence*

$$(M \cup h^0) \cup h^1 \cong M.$$

This is the handle-theoretic version of joining a newly created ball back to the old manifold and then cancelling the unnecessary creation.

(3) ***Cancelling inside a larger decomposition.*** *Suppose a 3-manifold is presented with two 0-handles, three 1-handles, and one 2-handle. Friedl's example first cancels one 0-handle with one 1-handle. Then the 2-handle cancels one of the remaining 1-handles. The result is a decomposition with one 0-handle and one 1-handle, hence an orientable solid torus $B^2 \times S^1$.*

(4) ***Creating a cancelling pair.*** *The reverse move is also important: for any boundary component $F \subset \partial M$ and any $k \in \{0, \dots, n-1\}$, one can introduce a cancelling pair consisting of a k-handle and a $(k+1)$-handle attached along F. This does not change the diffeomorphism type, but it gives extra handles that can later be used to cancel other handles.*

(5) ***Why handle slides enter.*** *A 2-handle whose attaching circle has algebraic intersection 3 with the belt sphere of a 1-handle cannot cancel that 1-handle by itself; nor can one with algebraic intersection -2. However, a suitable handle slide can replace these attaching data by a new attaching circle whose algebraic intersection is $3 - 2 = 1$. Such a new 2-handle is then a candidate for cancellation. This is the geometric reason that algebraic operations on boundary matrices are implemented by handle slides and cancellations.*

These examples also explain the main proof strategy in this lecture: create or modify a handle so that the attaching sphere has exactly one geometric intersection with the appropriate belt sphere, and then apply handle cancellation.

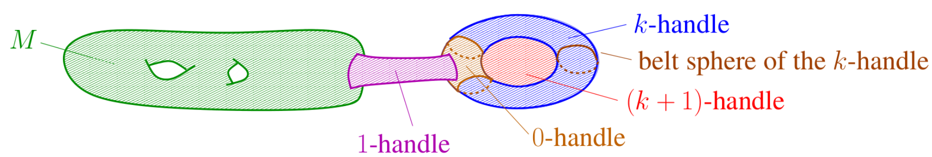

Figure 2: Friedl's picture of creating a cancelling pair. In the proof below, this move is used in the opposite direction: we create a new $(q+1)$–$(q+2)$ pair, use the new $(q+1)$-handle to cancel an old q-handle, and keep the newly created $(q+2)$-handle.

4 The elimination lemma: trading a handle upward

The next tool explains how a q-handle can be eliminated at the cost of introducing a new $(q+2)$-handle. This is the key operation in the proof.

Assume that the handlebody decomposition has no handles of index below q:

$$W \cong M_0 \times [0,1] + \sum_{i=1}^{p_q} h_i^q + \sum_{i=1}^{p_{q+1}} h_i^{q+1} + \sum_{i=1}^{p_{q+2}} h_i^{q+2} + \cdots .$$

Choose one q-handle, say $h_{i_0}^q$. Suppose we can find an embedding

$$\phi : S^q \times D^{n-q-1} \longrightarrow \partial_1 W_q$$

with the following two properties:

(i) The core sphere $\phi(S^q \times \{0\})$ is isotopic in $\partial_1 W_q$ to an embedding meeting the transverse sphere of $h_{i_0}^q$ exactly once and missing the transverse spheres of all other q-handles.

(ii) After the existing $(q+1)$-handles have been attached, the same core sphere becomes isotopic to a trivial embedding in $\partial_1 W_{q+1}$.

Then the chosen q-handle can be eliminated at the cost of adding one new $(q+2)$-handle.

Lemma 4.1 (Elimination lemma)**.** *Under the assumptions above, W is diffeomorphic relative to M_0 to a handlebody decomposition in which the handle $h^q_{i_0}$ is removed and one new $(q+2)$-handle is added. All other handles are unchanged up to isotopy.*

Idea of the proof. Because the embedding becomes trivial after the existing $(q+1)$-handles are attached, we may create a cancelling pair consisting of a new $(q+1)$-handle and a new $(q+2)$-handle. Creating a cancelling pair changes nothing diffeomorphically.

Now use the first property. The attaching sphere of the newly created $(q+1)$-handle can be arranged to meet the belt sphere of the chosen q-handle $h^q_{i_0}$ exactly once and to miss the belt spheres of all other q-handles. By the cancellation lemma, this new $(q+1)$-handle cancels $h^q_{i_0}$. After that cancellation, the only remaining trace of the operation is the newly created $(q+2)$-handle. Thus we have traded one q-handle for one $(q+2)$-handle.

The elimination lemma reduces the main problem to a construction problem. Given a q-handle, how do we construct the embedding ϕ required by the lemma? The answer uses three ingredients:

1. the acyclicity of the relative chain complex of (W, M_0);
2. the ability to modify an attaching sphere by adding boundaries of existing $(q+1)$-handles;
3. the Whitney trick, which removes algebraically cancelling intersection pairs.

5 Removing 0- and 1-handles

Before the main induction begins, we remove handles of index 0 and 1. For an h-cobordism, the inclusion

$$M_0 \hookrightarrow W$$

is a homotopy equivalence. In particular, it is 1-connected: it induces a bijection on path components and a surjection on fundamental groups.

Lemma 5.1 (No 0- and 1-handles)**.** *Let W be an n-dimensional compact cobordism with $n \geq 6$. If the inclusion $M_0 \hookrightarrow W$ is 1-connected, then W admits a handlebody decomposition relative to M_0 with no handles of index 0 or 1.*

Sketch. First consider a 0-handle. Since $M_0 \to W$ is bijective on path components, this 0-handle must be connected to the old boundary by some 1-handle. The core of that 1-handle gives the required single transverse intersection, so the 0-handle and the 1-handle cancel.

Now consider a 1-handle. To cancel it, we need a suitable 2-handle attaching circle meeting its belt sphere once. The surjectivity of

$$\pi_1(M_0) \longrightarrow \pi_1(W)$$

allows us to close a path through the 1-handle by a path near the old boundary in such a way that the resulting loop is nullhomotopic after the appropriate higher handles are attached. Since $n \geq 6$, a nullhomotopy disk can be arranged to be embedded by general position. Its normal bundle is trivial after choosing a framing, so the loop thickens to the attaching region of a 2-handle. The elimination lemma then removes the chosen 1-handle, possibly creating a 3-handle. Repeating this process removes all 1-handles.

For an h-cobordism, the hypotheses of the lemma are automatically satisfied. Thus, from now on, we may assume that the handlebody decomposition begins in index 2:

$$W \cong M_0 \times [0,1] + \sum_i h_i^2 + \sum_i h_i^3 + \cdots .$$

6 The relative chain complex from handles

Let

$$\pi := \pi_1(W) \cong \pi_1(M_0).$$

Passing to universal covers gives a cellular chain complex

$$C_*(\widetilde{W}, \widetilde{M}_0)$$

of free left $\mathbb{Z}[\pi]$-modules. The chosen handle decomposition supplies a basis: each r-handle gives one basis element in $C_r(\widetilde{W}, \widetilde{M}_0)$, after choosing a lift. We denote the basis element corresponding to h_i^r by

$$[h_i^r] \in C_r(\widetilde{W}, \widetilde{M}_0).$$

Since $M_0 \hookrightarrow W$ is a homotopy equivalence, the pair (W, M_0) is acyclic. Equivalently,

$$H_*(\widetilde{W}, \widetilde{M}_0) = 0.$$

Therefore the chain complex $C_*(\widetilde{W}, \widetilde{M}_0)$ is acyclic.

This acyclicity is the algebraic reason why lower handles can be eliminated. Suppose there are no handles below index q. Then

$$C_{q-1}(\widetilde{W}, \widetilde{M}_0) = 0.$$

Therefore the differential

$$d_q : C_q \longrightarrow C_{q-1}$$

is zero. Since the homology in degree q is zero, we get

$$\ker d_q = C_q = \operatorname{im} d_{q+1}.$$

Thus

$$d_{q+1} : C_{q+1} \longrightarrow C_q$$

is surjective. Consequently, for any chosen q-handle $h_{i_0}^q$, its basis vector can be written as

$$[h_{i_0}^q] = \sum_{j=1}^{p_{q+1}} x_j d_{q+1}([h_j^{q+1}])$$

for suitable coefficients $x_j \in \mathbb{Z}[\pi]$.

The next step is to turn this algebraic equality into a geometric attaching sphere for a new $(q+1)$-handle.

Remark 6.1. *The equality above is the central algebraic input of the proof. It says that every q-handle is algebraically the boundary of a formal $\mathbb{Z}[\pi]$-linear combination of $(q+1)$-handles. The modification lemma will realize this formal combination geometrically.*

7 The homology lemma and the role of the Whitney trick

Let $2 \leq q \leq n-3$. Consider an embedding

$$f : S^q \hookrightarrow \partial_1 W_q.$$

Assume that f is transverse to the transverse spheres of the q-handles. Since

$$\dim S^q + \dim S^{n-q-1} = q + (n-q-1) = n-1 = \dim(\partial_1 W_q),$$

the intersections are isolated points.

Each intersection point contributes a signed group element to the chain

$$[\widetilde{f}] \in C_q(\widetilde{W}, \widetilde{M}_0).$$

The signs record local intersection numbers, and the group elements record how paths from the base point wind around $\pi_1(W)$. Thus $[\widetilde{f}]$ is an algebraic intersection number of $f(S^q)$ with the transverse spheres of the q-handles.

$$\text{algebraic intersection number of } X \text{ and } Y \;:=\; X \cdot Y \;:=\; \sum_{P \in X \cap Y} \operatorname{sign}(P) \in \mathbb{Z}.$$

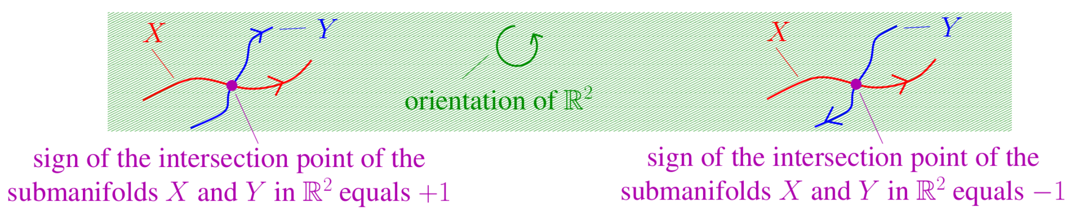

Figure 3: Friedl's sign convention for algebraic intersection numbers. The homology lemma uses the same idea, but with the extra group-ring label recording how each intersection point sits in the universal cover.

Lemma 7.1 (Homology lemma)**.** *Let $2 \leq q \leq n-3$ and $n \geq 6$. Let $f : S^q \hookrightarrow \partial_1 W_q$ be an embedding. Then f is isotopic to an embedding that meets the transverse sphere of $h^q_{i_0}$ exactly once and is disjoint from the transverse spheres of all other q-handles if and only if, after choosing a lift,*

$$[\widetilde{f}] = \pm g[h^q_{i_0}]$$

for some $g \in \pi$.

Idea of the proof. The implication from geometry to algebra is immediate. One transverse intersection with the belt sphere of $h^q_{i_0}$ contributes one signed group element, so the corresponding chain is $\pm g[h^q_{i_0}]$.

For the converse, put f in general position with respect to all transverse spheres. The equality

$$[\widetilde{f}] = \pm g[h^q_{i_0}]$$

says that all extra intersection points cancel algebraically in pairs: two points have opposite signs and represent the same element of the fundamental group. This is exactly the situation where the Whitney trick applies. Assuming the Whitney trick, each such pair can be removed by an isotopy of f, without creating new unwanted intersections. Repeating this process leaves precisely one intersection with the transverse sphere of $h^q_{i_0}$ and no intersections with the other transverse spheres.

Definition 7.2 (Whitney trick / W-trick, working form)**.** *Let M be an oriented smooth manifold, and let $X, Y \subset M$ be proper oriented submanifolds with*

$$\dim X + \dim Y = \dim M,$$

intersecting transversely. A pair of intersection points $P, Q \in X \cap Y$ is an algebraically cancelling pair *if P and Q have opposite local intersection signs and the loop obtained by joining P to Q along X and Q to P along Y is null-homotopic in the relevant complement. A* Whitney disk *for such a pair is an embedded disk whose boundary is the union of these two arcs and whose interior is disjoint from the relevant submanifolds.*

The Whitney trick *is the operation of using a framed Whitney disk to isotope one submanifold, usually Y, across the disk and thereby remove the two intersection points P and Q without changing the rest of the intersection data. In high-dimensional situations with enough room, for example when the dimensions and fundamental-group hypotheses in the Whitney Trick Theorem are satisfied, this operation can be repeated so that the number of geometric intersections is reduced to the absolute value of the algebraic intersection number.*

In this lecture the ambient manifold is $\partial_1 W_q$, the submanifolds are the attaching sphere $f(S^q)$ and the belt spheres of the q-handles, and the output is an isotopy of f that removes all algebraically cancelling extra intersections.

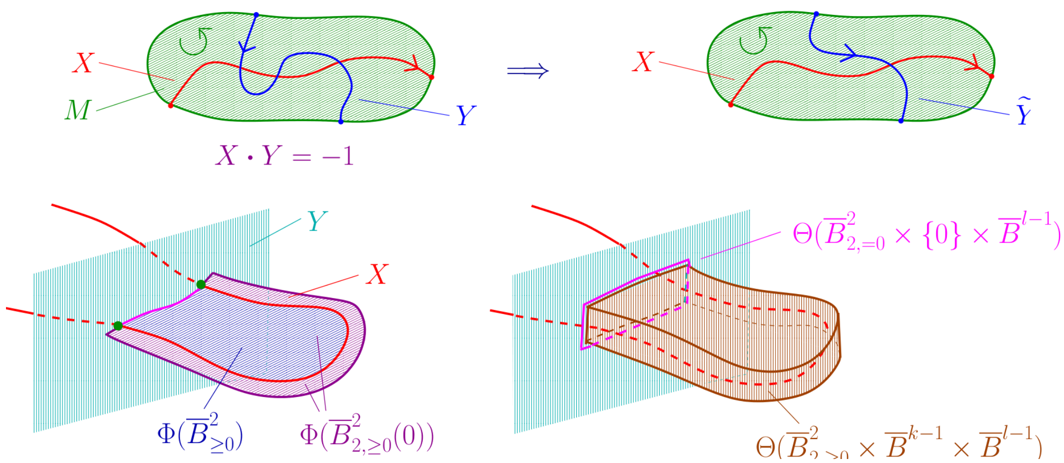

Figure 4: Friedl's pictures of the Whitney trick. The top picture shows the effect: after the isotopy, the unnecessary geometric intersections are removed while the algebraic intersection number is preserved. The lower picture illustrates the Whitney disk and the local push used to isotope one submanifold across it.

This is the only genuinely high-dimensional geometric input in the argument. The Whitney trick requires enough room to find Whitney disks and to use them without creating new intersections. Here the ambient boundary has dimension

$$\dim(\partial_1 W_q) = n - 1 \geq 5,$$

which explains the dimension hypothesis $n \geq 6$. The inequality $q \leq n - 3$ also ensures that the spheres involved have enough codimension for the required embedded and framed constructions.

Lecture comment. The homology lemma is the bridge from algebra to geometry. Algebra tells us which intersection points should cancel. The Whitney trick performs the actual geometric cancellation. Without the Whitney trick, the equality in the chain complex would not be enough.

8 The modification lemma

We still need to construct an attaching sphere whose algebraic intersection is exactly the desired basis vector $[h^q_{i_0}]$. The modification lemma provides this construction.

Start with a trivial embedding

$$\alpha : S^q \hookrightarrow \partial_1 W_q.$$

Since it is trivial, its class in the relative chain complex is zero:

$$[\widetilde{\alpha}] = 0.$$

The modification lemma says that we may alter α by connected summing it with attaching spheres of the existing $(q+1)$-handles. This changes its class by adding arbitrary $\mathbb{Z}[\pi]$-linear combinations of the boundaries of those $(q+1)$-handles, while keeping it isotopic to the original trivial embedding after the $(q+1)$-handles have been attached.

Lemma 8.1 (Modification lemma). *Let $f : S^q \hookrightarrow \partial_1 W_q$ be an embedding. For any coefficients*

$$x_1, \ldots, x_{p_{q+1}} \in \mathbb{Z}[\pi],$$

there is an embedding $g : S^q \hookrightarrow \partial_1 W_q$ such that:

(i) g is isotopic to f inside $\partial_1 W_{q+1}$;

(ii) after choosing compatible lifts,

$$[\widetilde{g}] = [\widetilde{f}] + \sum_{j=1}^{p_{q+1}} x_j d_{q+1}([h^{q+1}_j]).$$

Idea of the proof. It is enough to add one term at a time. Take the attaching sphere of a $(q+1)$-handle and form a connected sum with f along an embedded tube in $\partial_1 W_q$. This connected sum is isotopic to f once the corresponding $(q+1)$-handle has been attached, because the attaching sphere bounds the core disk of that handle. On the chain level, the operation adds the boundary of that $(q+1)$-handle.

A coefficient in $\mathbb{Z}[\pi]$ is a finite sum of signed group elements. By choosing the tube to wind around a loop representing an element of π, and by choosing the orientation of the connected-sum operation, one obtains a prescribed term $\pm g d_{q+1}([h^{q+1}_j])$. Repeating the construction for all terms in all coefficients gives the stated formula.

Combining the acyclicity of the chain complex with the modification lemma, we can now construct an embedding satisfying the hypotheses of the elimination lemma.

9 Induction: eliminating handles below a chosen index

We now prove the first half of the normal form lemma.

Proposition 9.1. *Let $(W; M_0, M_1)$ be an h-cobordism of dimension $n \geq 6$. Fix an integer q with $2 \leq q \leq n-3$. Then W admits a handlebody decomposition relative to M_0 with no handles of index $< q$.*

Proof. By Lemma 5.1, we may begin with a handlebody decomposition with no 0- or 1-handles. This proves the statement for $q = 2$.

Assume inductively that we have arranged that there are no handles of index $< q$, and suppose we want to remove the q-handles in order to obtain a decomposition with no handles of index $< q+1$. Choose one q-handle $h^q_{i_0}$.

Since there are no handles below index q, the chain group C_{q-1} is zero. The acyclicity of $C_*(\widetilde{W}, \widetilde{M}_0)$ therefore implies that

$$d_{q+1} : C_{q+1} \longrightarrow C_q$$

is surjective. Hence there are coefficients $x_j \in \mathbb{Z}[\pi]$ such that

$$[h^q_{i_0}] = \sum_j x_j d_{q+1}([h^{q+1}_j]).$$

Start with a trivial embedding

$$\alpha : S^q \hookrightarrow \partial_1 W_q.$$

By Lemma 8.1, there is an embedding

$$\beta : S^q \hookrightarrow \partial_1 W_q$$

which is isotopic to α inside $\partial_1 W_{q+1}$ and satisfies

$$[\widetilde{\beta}] = [\widetilde{\alpha}] + \sum_j x_j d_{q+1}([h^{q+1}_j]) = [h^q_{i_0}].$$

By Lemma 7.1, and using the Whitney trick, β is isotopic inside $\partial_1 W_q$ to an embedding meeting the transverse sphere of $h^q_{i_0}$ exactly once and missing all other transverse spheres of q-handles.

Thus the embedding β satisfies the two hypotheses of the elimination lemma: it has the correct single-intersection behavior before the $(q+1)$-handles are attached, and it becomes trivial after those handles are attached. Therefore Lemma 4.1 removes the chosen handle $h^q_{i_0}$ at the cost of one new $(q+2)$-handle.

Repeating this finitely many times removes all q-handles. This completes the induction step. Therefore, for the chosen original index q, all handles of index $< q$ can be eliminated. $\square$

The slogan is:

below the middle dimension, a q-handle can be pushed up to index $q+2$.

More precisely, the operation removes one q-handle and adds one $(q+2)$-handle. The total number of handles need not decrease, but the lowest index present in the decomposition increases.

10 Dual handles

The previous proposition eliminates low-index handles. To eliminate high-index handles, we use duality.

Suppose first that

$$W \cong M_0 \times [0,1] + h^r$$

has a single r-handle relative to M_0. If we reverse the cobordism and view W as built from M_1, then the same geometric piece is a handle of index

$$n - r.$$

This is the dual handle. Geometrically, the core and cocore exchange roles:

$$D^r \times D^{n-r} \quad \longleftrightarrow \quad D^{n-r} \times D^r.$$

For a full handlebody decomposition, the conclusion is that a decomposition relative to M_0 with p_r handles of index r gives a dual decomposition relative to M_1 with p_r handles of index $n - r$. Symbolically,

$$r \longleftrightarrow n - r.$$

Since $(W; M_0, M_1)$ is an h-cobordism, the reversed cobordism $(W; M_1, M_0)$ is also an h-cobordism. Hence the lower-elimination argument also applies to the dual decomposition.

Lecture comment. Duality is the reason we do not need a second independent argument for high-index handles. High-index handles in the original decomposition become low-index handles in the reversed decomposition.

11 Proof of the normal form lemma

We now prove Theorem 1.1.

Proof of the normal form lemma. Fix q with

$$2 \leq q \leq n - 3.$$

First apply Proposition 9.1 to the handlebody decomposition relative to M_0. We obtain a decomposition with no handles of index $< q$. Thus all remaining handles have index

$$q, q + 1, q + 2, \ldots, n.$$

It remains to remove the handles of index $> q + 1$.

Consider the dual handlebody decomposition relative to M_1. In the dual decomposition, an original handle of index r becomes a handle of index $n - r$. Therefore original handles of index

$$r \geq q + 2$$

correspond exactly to dual handles of index

$$n - r \leq n - q - 2.$$

Now apply Proposition 9.1 to the dual h-cobordism with the integer

$$n - q - 1.$$

This is allowed because $2 \leq q \leq n - 3$, so

$$2 \leq n - q - 1 \leq n - 3.$$

The result is that the dual decomposition can be chosen to have no handles of index

$$< n - q - 1,$$

or equivalently no handles of index at most $n - q - 2$. Dualizing back, this says that the original decomposition has no handles of index at least $q+2$. Combined with the first step, the only possible remaining indices are q and $q + 1$.

Therefore

$$W \cong M_0 \times [0,1] + \sum_{i=1}^{p_q} h_i^q + \sum_{j=1}^{p_{q+1}} h_j^{q+1}$$

relative to M_0, as required. □

References

[Fri] Stefan Friedl. *Topology.* See Chapter 256, "Handle cancellations and handle slides," and Chapter 258, "The Whitney trick." Figures in this handout are cropped from these chapters."

[LM24]
Wolfgang Lueck and Tibor Macko. *Surgery Theory: Foundations.* Springer, Cham, 2024. See Chapter 2, Section 2.4, especially Lemmas 2.34–2.37.

[Hir76]
Morris W. Hirsch. *Differential Topology.* Graduate Texts in Mathematics, Vol. 33. Springer, 1976.

[Kos93]
Antoni A. Kosinski. *Differential Manifolds.* Academic Press, 1993.